Managing BigData
Group Project Report

Team 7
Villanelle Chen, Chelsea Yeh
John Lee, Guilherme Felitti

1. Executive summary

Amazon's large-scale fulfillment network depends on clean, well-structured data to support ordering, shipping, and customer experience. To emulate this environment, our system integrates synthetic customer, order, and shipment data with real Amazon product metadata from public sources, creating a realistic yet fully de-identified dataset. A Third Normal Form relational database, complemented by NoSQL storage and a columnar warehouse, provides a consistent structure for both operational integrity and analytical workflows.

To address the limitations of Amazon's existing fulfillment tools, we developed a new application: **Shipment Delivery Risk Prediction Dashboard**. By integrating environmental, operational, and product attributes into a unified analytical table, an XGBoost model predicts shipment-level delivery risks before disruptions occur. A Streamlit interface enables real-time inference for operational use.

Insights from warehouse queries highlight where disruptions concentrate across facilities, carriers, product types, and seasonal patterns. These findings support proactively business decision-making, allowing Amazon to allocate resources earlier, improve delivery reliability, and enhance overall customer satisfaction.

---

2. Existing Systems

2.1 Dataset Resources

Our system combines synthetic operational tables with real product metadata sourced from public datasets. Customer, order, and shipment tables were fully generated using Python and Faker to avoid privacy concerns.

For the product, reviews, and session table, instead of fabricating all fields, we incorporated real Amazon product metadata from HuggingFace, Amazon Reviews as well as eCommerce behavior data from Kaggle to make them more realistic real-world business contexts. This hybrid approach makes the data structure and distributions more realistic while keeping all customer-related information synthetic and de-identified.

2.2 Database Design

To ensure structural consistency and data cleanliness, we designed a relational ER model that adheres to Third Normal Form. Core entities such as customers, products, orders, and shipments are organized into separate, nonredundant tables and linked through foreign keys and globally unique identifiers. This design minimizes duplication, maintains referential integrity across millions of records, and offers a stable foundation for downstream analytics. (The complete ER diagram is included in the Exhibits section.)

The relational database is supported by a NoSQL environment for semi-structured review text and session data, as well as a columnar data warehouse that enables efficient historical reporting, trend analysis, and the construction of machine-learning features.

---

3. New Business Application

To address the limitations of existing fulfillment systems that predominantly provide descriptive, post-hoc delivery updates, we propose a novel business application: **Shipment Delivery Risk Prediction Dashboard**.

This application introduces a predictive framework capable of estimating shipment-level delivery outcomes prior to the manifestation of service disruptions. The development of this system required constructing an expanded dataset that moves beyond transactional

shipment logs and incorporates a broader set of determinants known to influence logistical performance.

Specifically, the dataset integrates environmental variables (e.g., precipitation, wind speed, traffic congestion), operational indicators (e.g., carrier reliability metrics, warehouse load indices, route difficulty scores), and enriched customer and order attributes. These features were integrated within MySQL RDS to form a consolidated analytical table suitable for supervised multi-class classification.

In the modeling stage, XGBoost was chosen for its ability to capture complex feature interactions and handle class imbalance. The model achieved a macro-F1 score of about 0.52, providing enough predictive power to support operational triage and early identification of high-risk shipments.

To enable practical deployment, we built a Streamlit-based interface that operationalizes the model by connecting directly to the underlying RDS database and replicating the full preprocessing pipeline used during training. Upon input of a shipment identifier, the application retrieves relevant records, reconstructs required feature representation, and produces probabilistic predictions for four delivery risk categories, accompanied by recommended operational responses.

By delivering forward-looking risk assessments rather than retrospective updates, this application represents a valuable enhancement to Amazon's existing delivery-tracking tools. It provides a decision-support mechanism that enables earlier detection of emerging logistical issues, thereby improving fulfillment responsiveness and supporting more efficient exception management across the network.

---

4. Decision-Making and Insights

The analytical warehouse enables a level of strategic visibility that cannot be achieved through operational logs alone. By organizing original datasets and model predictions into a star schema, the system supports multi-dimensional aggregation that reveals patterns across three main areas. (Query results have been attached to the Exhibit section.)

4.1 Operational Insight

Order outcomes are evenly distributed across all status categories, with each state (returned, shipped, delivered, cancelled, processing, pending) representing roughly 16–17% of total orders. This balanced pattern suggests a stable fulfillment pipeline with no single stage showing disproportionate congestion or failure rates.

As for Carrier Performance, across major carriers—USPS, FedEx, Amazon Logistics, DHL, and UPS—average delivery delays remain tightly clustered at around 2.13 days. Although total shipment volumes vary, each carrier shows a similar number of delayed shipments, indicating consistent performance across logistics partners and no clear outlier in service reliability.

4.2 Customer Insight

Customer distribution is concentrated in a few key markets, with Korea and Congo showing the largest user bases, followed by Bulgaria, Bolivia, and Romania. This indicates a strong international reach across diverse regions.

Registration trends show rapid growth after 2020, with annual new customers stabilizing around 190,000–200,000 through 2024. Although 2025 shows a slight dip, overall customer acquisition remains strong, reflecting continued platform expansion.

4.3 Environmental and Temporal Insight

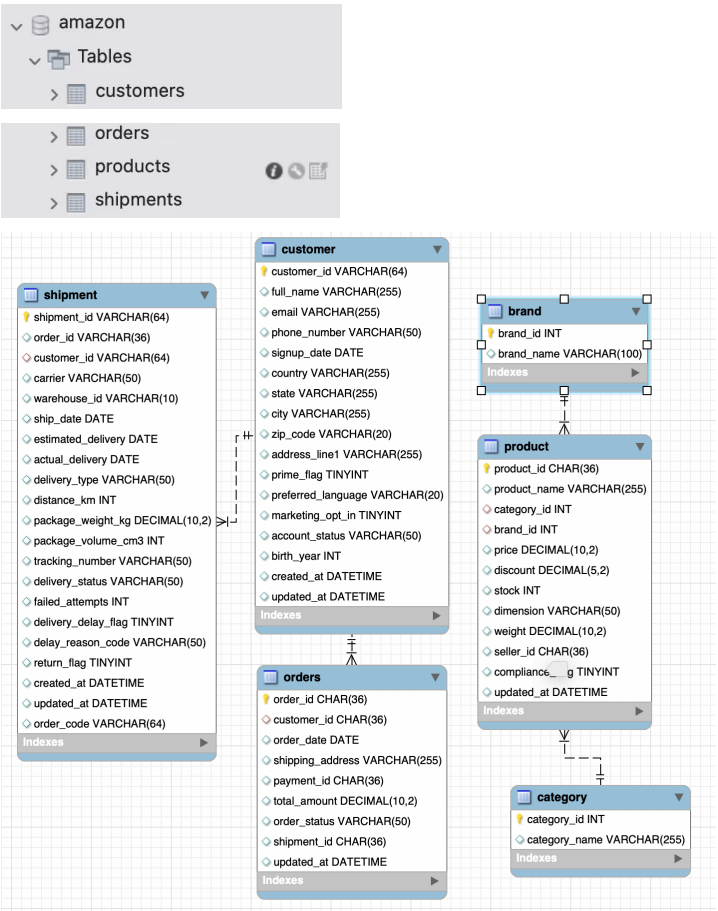
Shipment performance shows a slightly higher delay rate on holidays (0.718) compared to non-holiday periods (0.712). Although the difference is small, it reflects the added strain that seasonal demand places on the fulfillment network.

Across regions, higher average rainfall is associated with marginally longer delivery times, with delays ranging from 2.12 to 2.14 days in areas such as PHX3, SFO5, and SEA1. These patterns suggest that weather conditions contribute modest but measurable impacts on delivery speed.

---

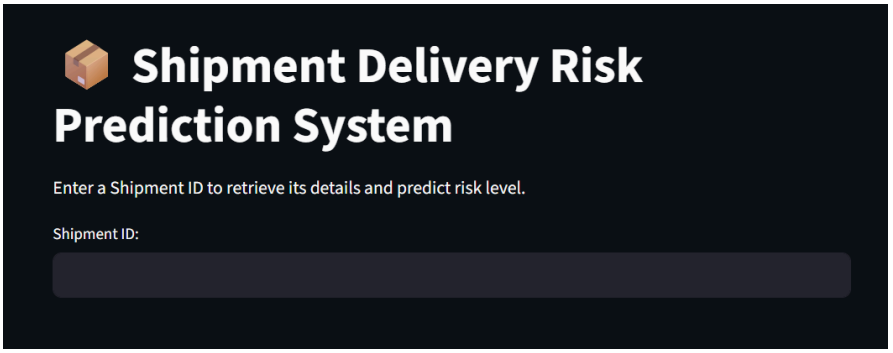
5. Exhibits

5.1 ER diagrams

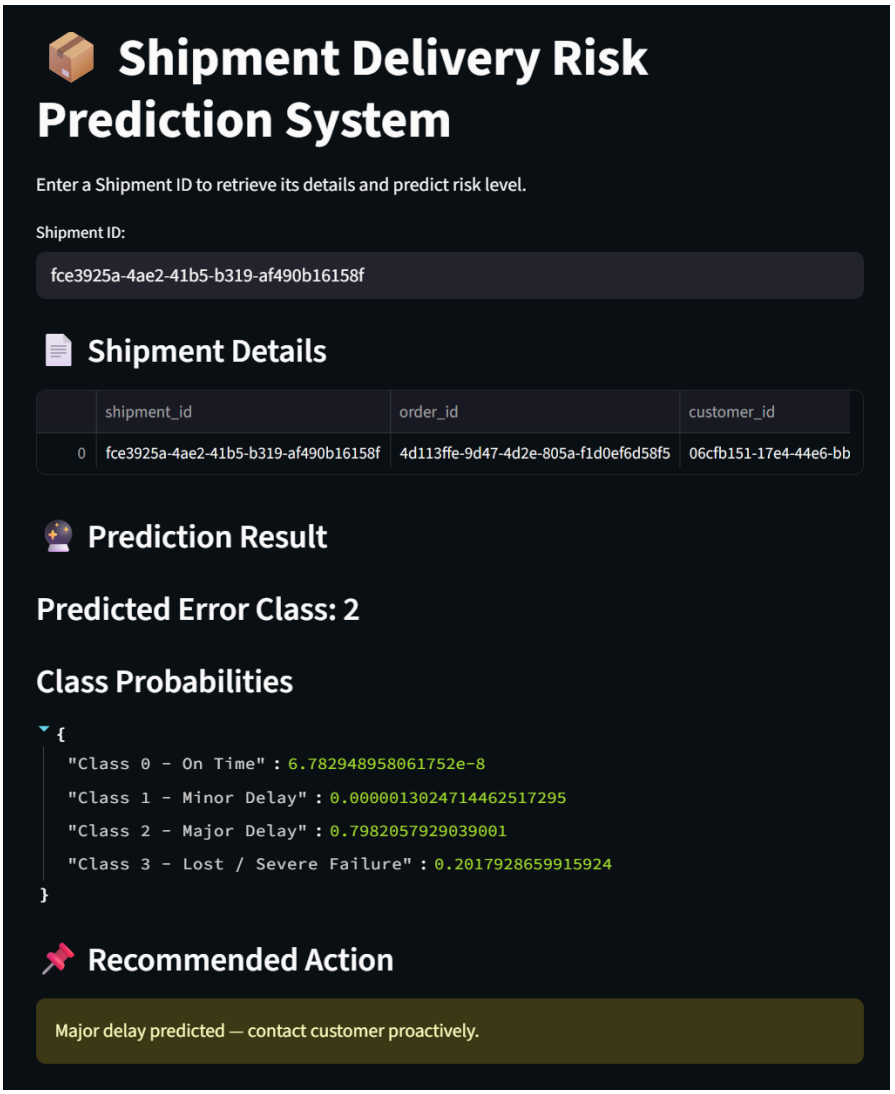


5.2 Demonstration of Real-Time Shipment Risk Inference via Streamlit UI (Link)

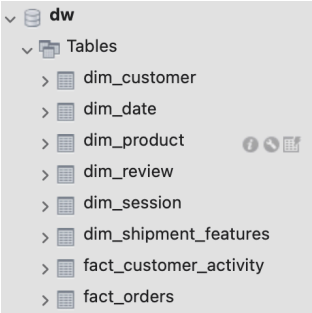
5.2.1 Initial Page of the Application

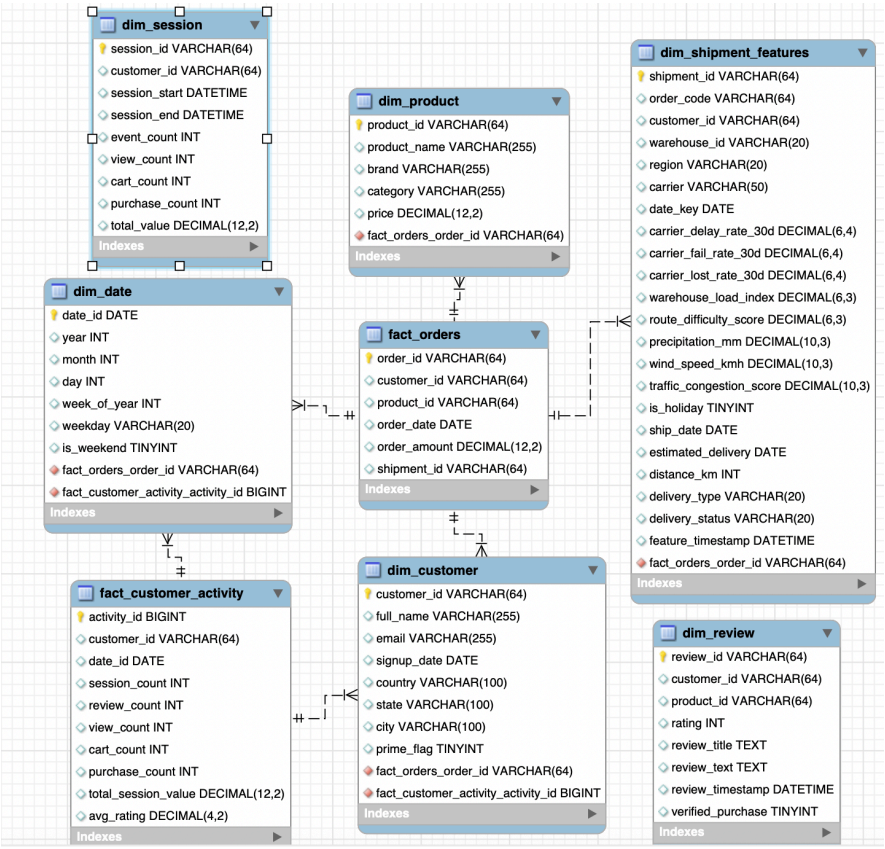


5.2.2 Result Page of the Application



5.3 RDS-SQL-SCHEMA





5.4 Query Results

5.4.1 Operational Insights

Order status

Order_status	num_orders	pct_orders
returned	334078	16.70
shipped	334021	16.69
delivered	333892	16.69
cancelled	333322	16.66
processing	332897	16.64
pending	332790	16.63

Carrier

carrier	avg_delay_days	delayed_shipments	total_shipments
USPS	2.1332	284590	399392
FedEx	2.1319	284478	399709
Amazon Logistics	2.1311	284995	400023
DHL	2.1309	284977	400102
UPS	2.1281	284866	400774

5.4.2 Customer Insights

Customer Countries Distribution (Top 5)

country	customer_count
Korea	8139
Congo	8134
Bulgaria	4242
Bolivia	4206
Romania	4204

Yearly Registration Trends

year	new_customers
2020	16369
2021	199931
2022	199844
2023	199820
2024	200166
2025	183870

5.4.3 Environmental and Temporal Insights

Holiday Influence

is_holiday	shipments_processed	delay_rate
0	1983615	0.712
1	16385	0.718

Precipitation Speed Influence

region	avg_rain	avg_delay_days
PHX3	3.2	2.14
SFO5	3.25	2.14
ORD9	3.12	2.13
DFW7	3.3	2.13
EWR4	3.23	2.13
SEA1	3.34	2.13
ONT8	3.07	2.12